# A Project Report on
# E-Voting System


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# SUPERVISOR RECOMMENDATION

This is to certify that this project entitled, “E-Voting System”, prepared and submitted by Aayusha Neupane and Siddhartha Siddhi Bajracharya in partial fulfilment of the requirements of the degree of Bachelor of Computer System and Information Technology (BCSIT) awarded by Pokhara University, has been completed under my supervision. I recommend the same for acceptance by the University.

……………….

Supervisor

Mr. Sanju Shrestha

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# CERTIFICATE OF APPROVAL

This project, entitled "E-Voting System", prepared and submitted by Aayusha Neupane and Siddhartha Siddhi Bajracharya, has been examined and is accepted for the award of the degree of Bachelor of Computer System and Information Technology (BCSIT) awarded by Pokhara University.

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# ABSTRACT

The E-Voting System is a web-based application developed to provide a secure, efficient, and user-friendly platform for conducting elections electronically. The system is designed using HTML, CSS, JavaScript, PHP, and MySQL. It enables registered voters to cast their votes online while ensuring that each voter can vote only once.

The system includes features such as voter authentication, party symbol management, vote casting. The administrator can manage voter and party information and monitor the election process. The application reduces the time and effort required in traditional voting systems and minimizes the possibility of human errors during vote counting.

The project demonstrates the practical application of web technologies and database management systems in developing a reliable electronic voting platform. The developed system provides transparency, accuracy, and efficiency in the election process and can be further enhanced with advanced security features in the future.

# ACKNOWLEDGEMENT

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# Abbreviations

| Abbreviation | Full Form |
|---|---|
| CSS | Cascading Style Sheets |
| CRUD | Create, Read, Update, Delete |
| DBMS | Database Management System |
| HTML | HyperText Markup Language |
| HTTP | HyperText Transfer Protocol |
| ID | Identification |
| JS | JavaScript |
| MySQL | My Structured Query Language |
| OTP | One-Time Password |
| PHP | Hypertext Preprocessor |
| SQL | Structured Query Language |
| UI | User Interface |
| UX | User Experience |

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# Chapter 1: Introduction

## 1.1.Background

Voting is one of the fundamental rights in a democratic system. Traditional paper-based voting systems require significant manpower, time, and resources. They are also prone to human errors during vote counting and result compilation. With advancements in information technology, electronic voting systems have emerged as an efficient alternative to conventional voting methods.

The E-voting system is a web based website which is designed to conduct political elections digitally in a secure, efficient, and transparent manner in Nepal. It allows the registered voters of the country to cast their votes while ensuring the confidentiality. The system replaces traditional physical voting systems with a modern solution which saves time, reduces the cost and minimizes errors.

## 1.2.Problem Statement

Currently, Voters face a lot of problems such as:

1.Walking far distances to vote

2.Hectic and tedious voting systems

Which leads to many citizens not being able to vote, or experiencing a hectic and tedious voting experience.

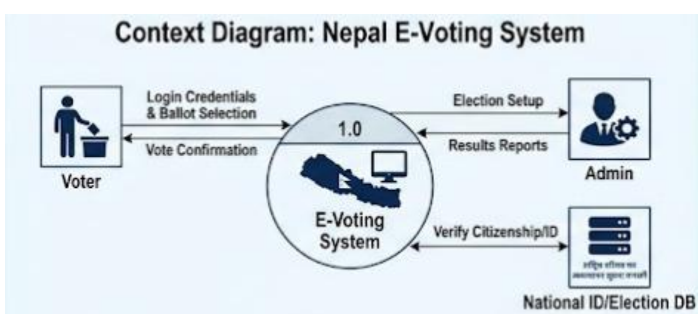


*Figure 1. Traditional Voting System of Nepal*

This project aims to create an online approach for voters through which a citizen can vote just a few clicks away.

## 1.3.Objectives

- To develop a secure online voting platform
- Ensure a transparent and fair election
- Save time for both the voters and the staff involved
- To prevent duplication and illegal voting.

## 1.4 Scope of the Project

The E-Voting System is designed for educational institutions, organizations, and small-scale elections. The system includes voter registration, authentication, party management, vote casting functionalities.

## 1.5 Significance of the Study

The system reduces paperwork, saves time, minimizes errors, and provides quick and accurate election results. It also enhances transparency and accessibility in the voting process.

# Chapter 2: Literature Review

## 2.1 Introduction

Electronic Voting(E-Voting) systems are digital platforms that allow elections to be conducted online instead of traditional paper-based methods. Traditional paper-based voting systems require significant manpower, time, and resources. They are also prone to human errors during vote counting and result compilation.

E-Voting allows the registered voters of the country to cast their votes while ensuring confidentiality. These systems aim to improve efficiency, transparency, accuracy and accessibility in the elections. The system replaces traditional physical voting systems with a modern solution which saves time, reduces the cost and minimizes errors.

## 2.2 Review of Existing Systems and Theories

Several countries have explored electronic voting systems to improve election efficiency and accessibility.

In 2005, Estonia became the first nation to implement nationwide internet voting. Estonian election officials declared the electronic voting system a success and found that it withstood the test of real-world use.

Research on e-voting systems shows that they can improve:

- Election efficiency
- Faster result generation
- Reduction of manual errors
- Better accessibility for voters

However, studies also identify challenges such as:

- Cybersecurity risks
- Infrastructure requirements

- Public trust in online systems

Switzerland's E-Voting Trials

- Switzerland has introduced e-voting through controlled trial programs instead of implementing it nationwide. The country follows a gradual approach to evaluate the security, reliability, and effectiveness of online voting before expanding its use.
- The primary objective of these trials is to improve voting accessibility, particularly for Swiss citizens living abroad and voters with visual impairments, who may face difficulties participating in traditional paper-based elections.
- In 2023, selected Swiss cantons began allowing eligible voters to participate in elections using an online voting system. This marked an important step toward providing secure and accessible digital voting for specific groups of citizens.
- One of the major advantages of the system is that it enables blind and visually impaired voters to cast their votes independently, without requiring assistance from another person. This helps ensure voter privacy, equality, and independence during the electoral process.
- Switzerland places strong emphasis on security, transparency, voter authentication, and continuous system evaluation throughout the trial period. These measures help identify potential vulnerabilities and improve the system before wider adoption.
- The successful implementation of these trials demonstrates that e-voting has the potential to increase voter accessibility and convenience while maintaining election integrity. Switzerland's cautious approach highlights the importance of extensive testing, strong cybersecurity measures, and public trust before deploying an online voting system on a national scale.

France's Online Voting for Citizens Living Abroad

- Since 2012, France has allowed French citizens living abroad to vote online in certain elections, making it easier for expatriates to participate in the democratic process without returning to France.
- The introduction of online voting has improved accessibility and convenience for overseas voters by enabling them to cast their votes remotely through a secure electronic platform.
- France adopted this approach to encourage greater voter participation among its citizens residing outside the country while reducing the logistical challenges associated with traditional voting methods.
- The system demonstrates how online voting can support electoral participation for specific groups of voters who may face geographical barriers to voting.
- France's experience highlights the potential of e-voting to improve accessibility while emphasizing the need for robust security measures and continuous evaluation to ensure the integrity and reliability of the electoral process.

In the context of Nepal, geographical barriers and limited digital infrastructure highlights the need for a simple and practical e-voting solution.

Therefore, this project focuses on developing a basic academic model of a web-based E-voting system emphasizing usability, transparency, and efficiency.

# Chapter 3: System Analysis & Design

## 3.1 System Analysis

System analysis is the process of studying the current system, identifying problems, and determining the requirements for the proposed system. The proposed E-Voting System aims to replace manual voting procedures with a secure and efficient web-based platform developed using PHP, HTML, CSS, JavaScript, and MySQL.

The system allows administrators to manage candidates and voters while enabling authenticated voters to cast their votes electronically. The system automatically counts votes and generates election results.

## 3.2 Requirement Analysis

Requirement analysis identifies the functionalities and constraints required for the successful operation of the system.

### 3.2.1 Functional Requirements

The E-Voting System shall provide the following functionalities:

**Administrator**

- Login to the system
- Add, edit, and delete party symbols
- Add and manage voter information
- Start and end elections

**Voter**

- Login using citizenship number and password
- View available party symbols
- Cast vote
- View voting confirmation

**Use Case List**

*Table 1. Use Case Table*

| Actor | Use Cases |
|---|---|
| Admin | Login, Manage party symbols, Manage Voters |
| Voter | Login, View party symbols, Cast Vote, Logout |

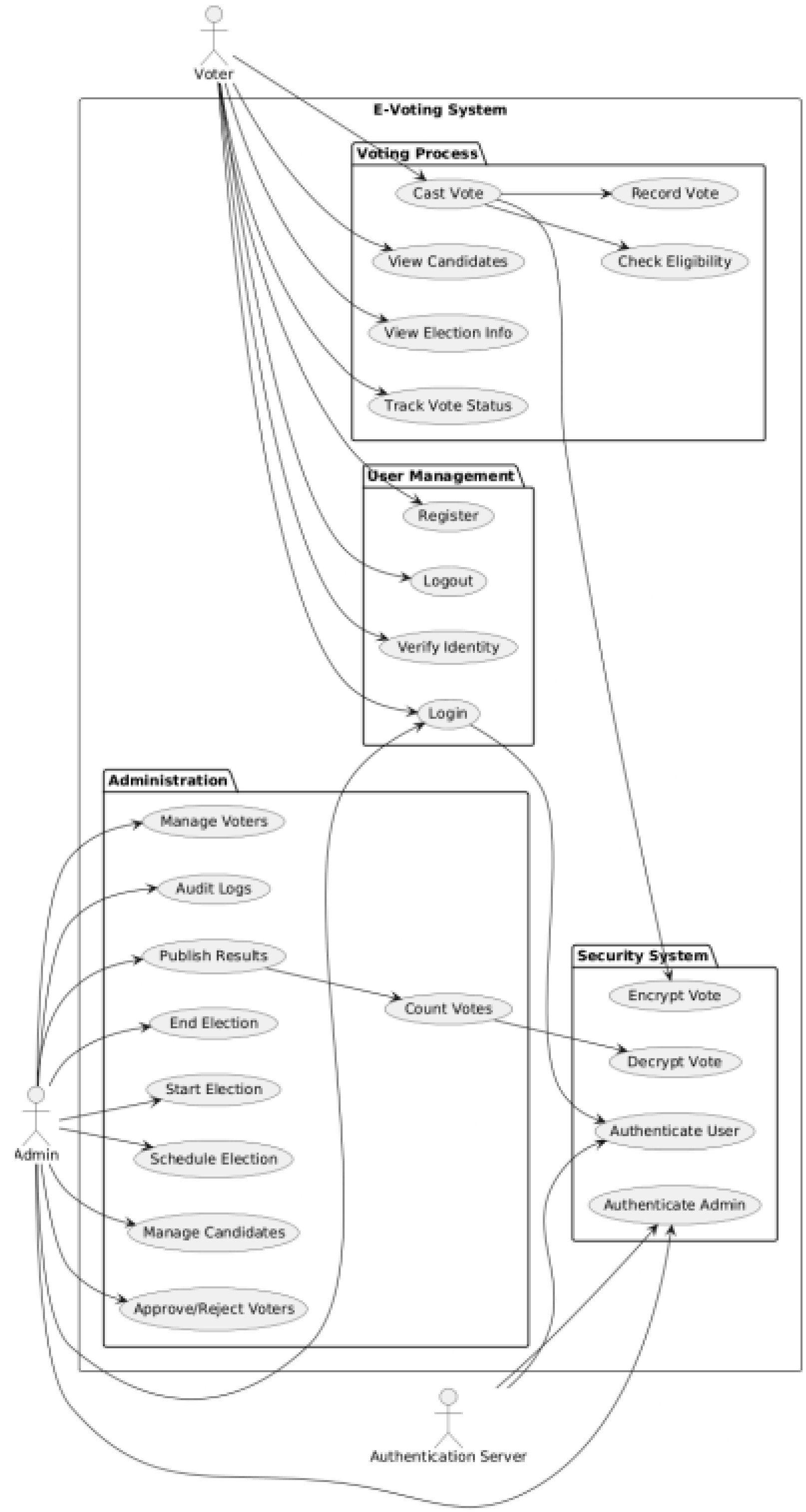


*Figure 2.Use Case Diagram*

### 3.2.2 Non-Functional Requirements

**Security**

- Only registered voters can vote.
- Passwords should be secured.
- Duplicate voting must be prevented.

**Reliability**

- The system should operate without data loss.
- Votes should be stored accurately.

**Performance**

- The system should respond quickly to user requests.
- Results should be generated automatically.

**Usability**

- The interface should be simple and user-friendly.
- Users should be able to navigate easily.

**Maintainability**

- The system should allow future updates and modifications.

## 3.3 Feasibility Analysis

### 3.3.1 Technical Feasibility

The project is technically feasible because the required technologies such as PHP, HTML, CSS, JavaScript, MySQL, and XAMPP are readily available and well-supported. The development team possesses sufficient knowledge of these technologies.

### 3.3.2 Operational Feasibility

The proposed system is easy to operate and requires minimal training. Users can easily access the voting platform through a web browser and perform their tasks efficiently.

#### 3.3.3 Financial Feasibility

The project is financially feasible because all development tools used are open-source and free. No significant investment is required apart from basic computer hardware and internet access.

#### 3.3.4 Schedule Feasibility

The project can be completed within the allocated academic project timeline. The development phases include planning, analysis, design, coding, testing, and documentation.

### 3.4 Data Modeling

The database consists of three major entities:

**Voters**

Attribute

voter_id

voter_name

citizenship_no

email

**Votes**

Attribute

voter_id

symbol_id

**Entity Relationships**

- One voter can cast only one vote.
- One party can receive multiple votes.
- Each vote belongs to one voter and one candidate.

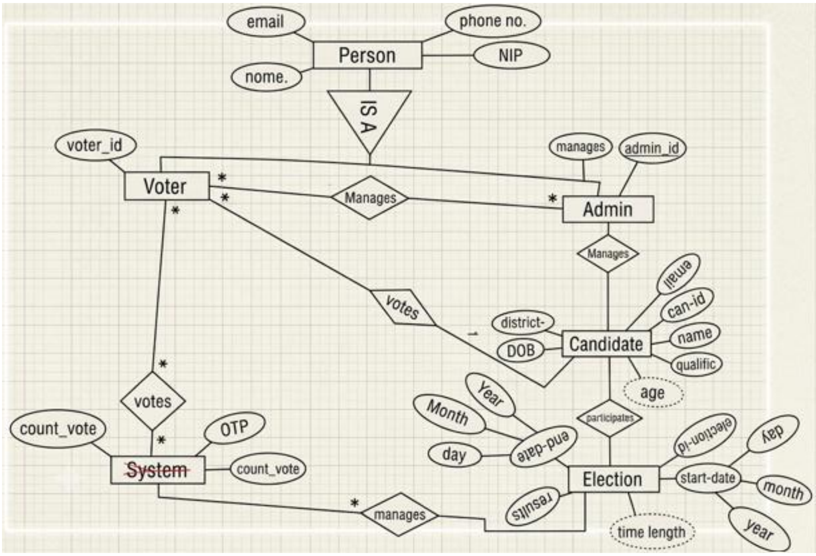


*Figure 3. ER Diagram*

## 3.5 Process Modeling

### Voting Process

1. Voter enters login credentials.
2. System verifies voter information.
3. Party symbol list is displayed.
4. Voter selects a party symbol.
5. Vote is recorded in the database.
6. System updates vote count.
7. Voting confirmation is displayed.

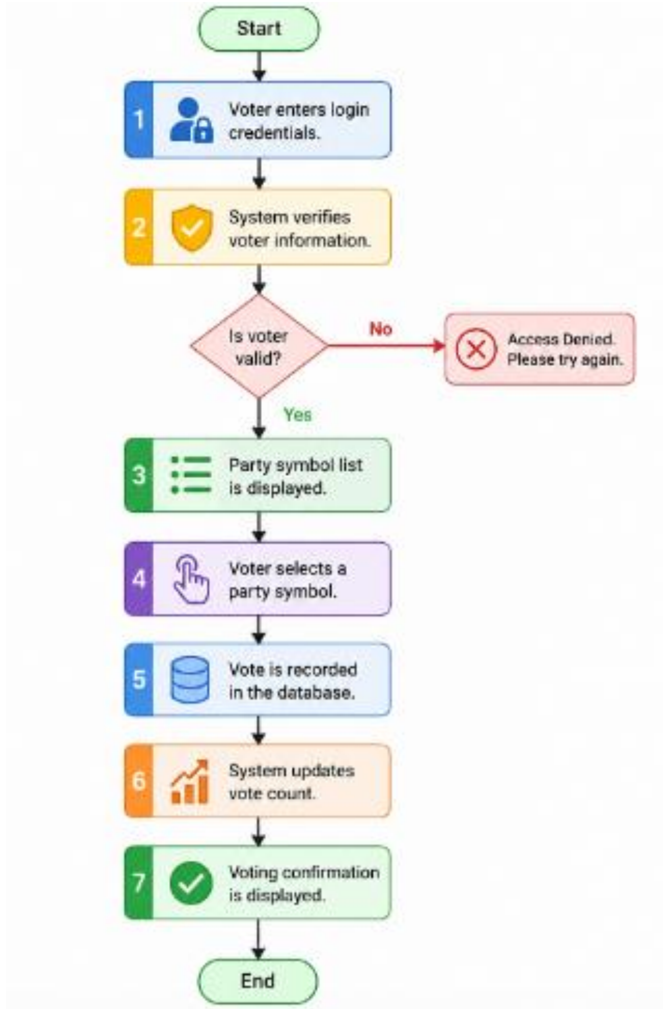


*Figure 4. Flowchart*

## 3.6 Architectural Design

The E-Voting System follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

### 3.6.1 Presentation Layer

The presentation layer is developed using HTML, CSS, and JavaScript. It provides the user interface through which voters interact with the system. This layer includes the Home Page, Guidelines Page, Login Page, Voting Page, Thank You Page, and Results Page.

### 3.6.2 Application Layer

The application layer is developed using PHP. It handles user authentication, vote validation, vote recording, result generation, and communication between the user interface and the database.

### 3.6.3 Database Layer

The database layer uses MySQL to store voter information, party information, and voting records. It ensures data consistency and secure storage of election data.


*Figure 5. System Architecture Diagram*

## 3.7 Database Schema Design

The E-Voting System uses three database tables: **users**, **symbols**, and **votes**. These tables store voter information, political party symbols, and voting records respectively.

**Table: users**

*Table 2. Users database table*

| **Field Name** | **Type** | **Description** |
|---|---|---|
| citizenship_no | VARCHAR(50) (PK) | Citizenship number of voter |
| full_name | VARCHAR(100) | Full name of voter |
| voter_id | VARCHAR(50) | Voter identification number |
| email | VARCHAR(100) | Email address of voter |

**Table: symbols**

*Table 3. Symbols Database Table*

| Field Name | Type | Description |
|---|---|---|
| symbol_id | INT (PK) | Unique symbol identifier |
| symbol_name | VARCHAR(100) | Name of political party |
| symbol_image | VARCHAR(255) | Party symbol image |

**Table: votes**

*Table 4. Votes Table*

| Field Name | Type | Description |
|---|---|---|
| citizenship_no | VARCHAR(50) (FK) | References users table |
| symbol_id | INT (FK) | References symbols table |

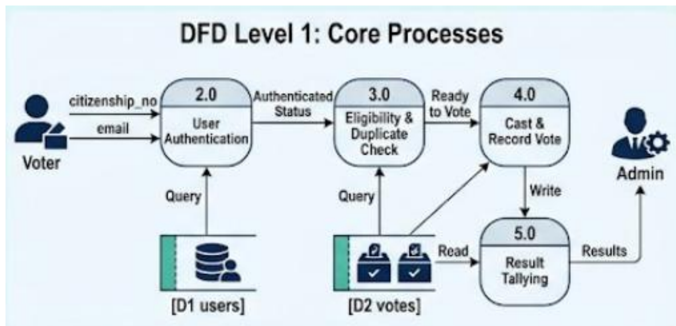


*Figure 6. Data Flow Diagram Level 1*

## 3.8 Interface Design

The user interface is designed to be simple, responsive, and user-friendly.

**Home Page**

Displays an introduction to the E-Voting System and provides navigation to the guidelines page.

**Guidelines Page**

Displays voting instructions and election rules. Users proceed by clicking the Continue button.

**Login Page**

Allows voters to enter their Full Name, Citizenship Number, Voter ID, and Email Address for verification.

**Voting Page**

Displays 27 party symbols. The voter selects one symbol to cast a vote.

**Vote Confirmation Dialog**

Prompts the voter to confirm or cancel the selected vote before submission.

**Thank You Page**

Displays a message confirming successful participation in the election.

## 3.9 Physical Model

The physical model describes how the system components are deployed and connected.

**Client Side**

- Web Browser
- HTML
- CSS
- JavaScript

**Web Server**

- Apache Server (XAMPP)

**Application Layer**

- PHP

**Database Server**

- MySQL

**Deployment Environment**

- Windows Operating System
- Localhost (XAMPP)

# Chapter 4: Implementation & Testing

## 4.1 Implementation

The E-Voting System was implemented as a web-based application using PHP, HTML, CSS, JavaScript, and MySQL. The system allows voters to view election guidelines, authenticate themselves using voter details, cast votes for political party symbols, and view election results. The application was developed following a modular approach to ensure maintainability and scalability.

## 4.2 Tools Used

**Programming Languages**

- HTML – Used to create the structure of web pages.
- CSS – Used for styling and responsive user interface design.
- JavaScript – Used for client-side validation, alerts, confirmations, and user interaction.
- PHP – Used for server-side processing and database interaction.

**Database Platform**

- MySQL – Used to store voter information, party symbols, and voting records.

**Web Server**

- XAMPP (Apache Server) – Used as the local development environment.

**Development Tools**

- Visual Studio Code – Source code editor.
- phpMyAdmin – Database administration tool.
- Brave – Browser used for testing and debugging.

**CASE Tools**

- Draw.io / Lucidchart – Used for ER Diagrams, Flowcharts, and System Architecture Diagrams.

## 4.3 Implementation Details of Modules

**Module 1: Home Page**

**Purpose:**
Provides an introduction to the E-Voting System.

**Functions:**

- Display welcome message.
- Navigate to Guidelines Page.

**Module 2: Guidelines Page**

**Purpose:**
Displays election rules and voting instructions.

**Functions:**

- Show voting guidelines.
- Redirect voter to Login Page through Continue button.

**Module 3: Login Module**

**Purpose:**
Collects voter information and validates user identity.

**Input Fields:**

- Full Name
- Citizenship Number

- Voter ID
- Email Address

**Functions:**

- Validate input fields.
- Verify voter information.
- Redirect voter to Voting Page.

**Module 4: Voting Module**

**Purpose:**
Allows voters to select a political party symbol.

**Functions:**

- Display 27 party symbols.
- Record selected symbol.
- Display vote confirmation dialog.

**Procedure:**

1. User selects a symbol.
2. Confirmation message appears.
3. If user confirms, vote is stored in database.
4. Success notification is displayed.

**Module 5: Vote Recording Module**

**Purpose:**
Stores voting information securely in the database.

**Functions:**

- Insert citizenship number and symbol ID into votes table.
- Prevent invalid data entries.

**Module 6: Thank You Module**

**Purpose:**

Displays vote submission acknowledgement.

**Functions:**

- Display thank you message.
- Provide View Results button.

## 4.4 Testing

Testing was performed to verify that all system functionalities operate correctly and meet project requirements.

### 4.4.1 Unit Testing

Unit testing was conducted on individual modules.

**Test Case 1: Login Validation**

*Table 5. Login Validation Test Table*

| **Test Item** | **Description** |
|---|---|
| Module | Login |
| Input | Valid voter details |
| Expected Result | User redirected to Voting Page |
| Actual Result | User redirected successfully |
| Status | Pass |

**Test Case 2: Empty Login Fields**

*Table 6. Login Validation Test Table*

| Test Item | Description |
|---|---|
| Module | Login |
| Input | Empty fields |
| Expected Result | Validation error message |
| Actual Result | Error displayed |
| Status | Pass |

**Test Case 3: Vote Submission**

*Table 7. Voting Test Table*

| Test Item | Description |
|---|---|
| Module | Voting |
| Input | Symbol Selection |
| Expected Result | Vote stored successfully |
| Actual Result | Vote stored successfully |
| Status | Pass |

# Chapter 5: Conclusion and Future Recommendations

## 5.1 Lessons Learnt

The development of the E-Voting System provided valuable practical experience in web application development and database management. Through this project, knowledge gained from academic courses was applied in a real-world scenario.

The key lessons learned include:

- Designing and developing dynamic web applications using PHP.
- Creating responsive user interfaces using HTML, CSS, and JavaScript.
- Managing and querying databases using MySQL.
- Implementing database connectivity between PHP and MySQL.
- Understanding software development life cycle (SDLC) concepts.
- Designing system models such as ER diagrams, flowcharts, and architecture diagrams.
- Performing testing and debugging to improve system reliability.
- Working with user authentication, data validation, and vote recording mechanisms.

## 5.2 Conclusion

The E-Voting System was successfully designed and implemented using PHP, HTML, CSS, JavaScript, and MySQL. The system provides a simple, secure, and efficient platform for conducting electronic voting.

The application allows voters to view election guidelines, submit their voter information, cast votes through party symbols, and view election results. The system automates vote recording and result generation, reducing manual effort and minimizing the possibility of human errors.

The project successfully achieved its objectives by providing a user-friendly voting environment while maintaining data integrity and efficient vote management.

## 5.3 Recommendations

Although the current system fulfills its intended objectives, several improvements can be made in future versions:

- Add encryption techniques to improve data security.
- Introduce biometric verification for voter identification.
- Develop a mobile application version of the system.
- Enable real-time result visualization using graphs and charts.
- Implement cloud-based deployment for wider accessibility.
- Add administrative controls for election scheduling and monitoring.
- Enhance system scalability to support larger elections and a greater number of voters.

## 5.4 Future Enhancement

As this project is an academic prototype, several advanced security features were beyond the scope of the current implementation. These features can be incorporated in future versions to strengthen voter authentication, improve system security, and enhance the reliability of the voting process.

- Government Digital Identity Integration: Integrate the system with a government-issued digital identity platform to verify voter identity before allowing access to the voting process.
- Biometric Authentication: Implement biometric verification, such as fingerprint or facial recognition, to ensure that only the legitimate voter can cast a vote.
- Trusted Device Registration: Allow voting only from registered or trusted devices, while requiring additional verification when login attempts are made from a new or unrecognized device.
- Digital Signatures: Use digital signatures to verify the authenticity and integrity of votes, ensuring that vote data cannot be altered after submission.
- Multi-Factor Authentication: Strengthen the existing email OTP verification by adding SMS OTP or other authentication methods to provide an additional layer of security.

# Appendices

Appedix A: User Interface Screenshots


*Figure 7.Home Page*

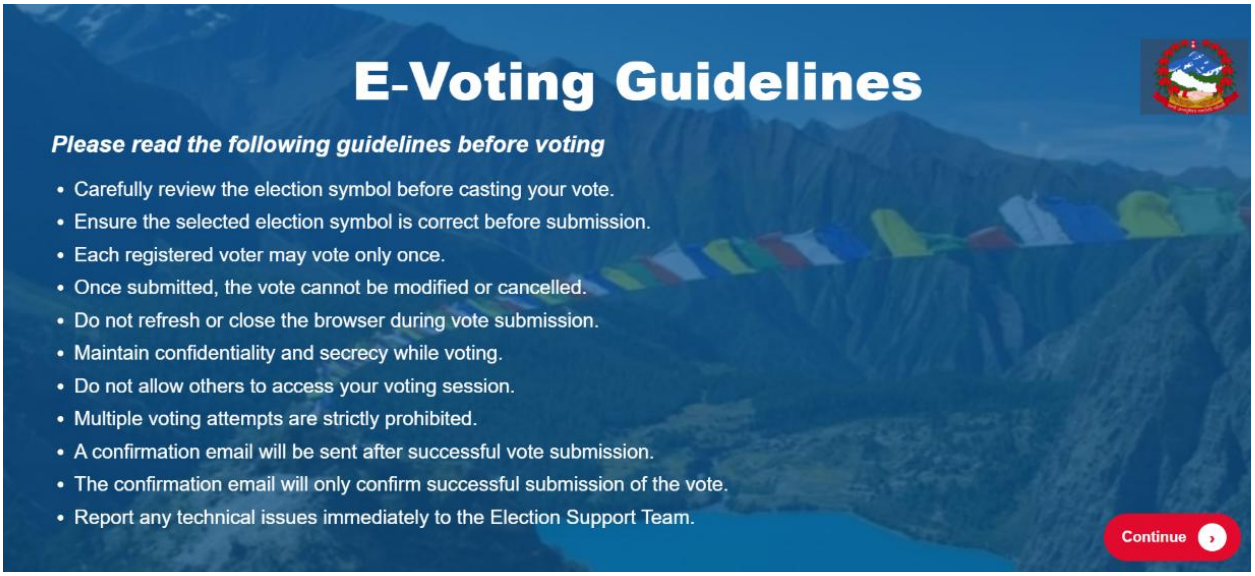


*Figure 8.Guideline Page*

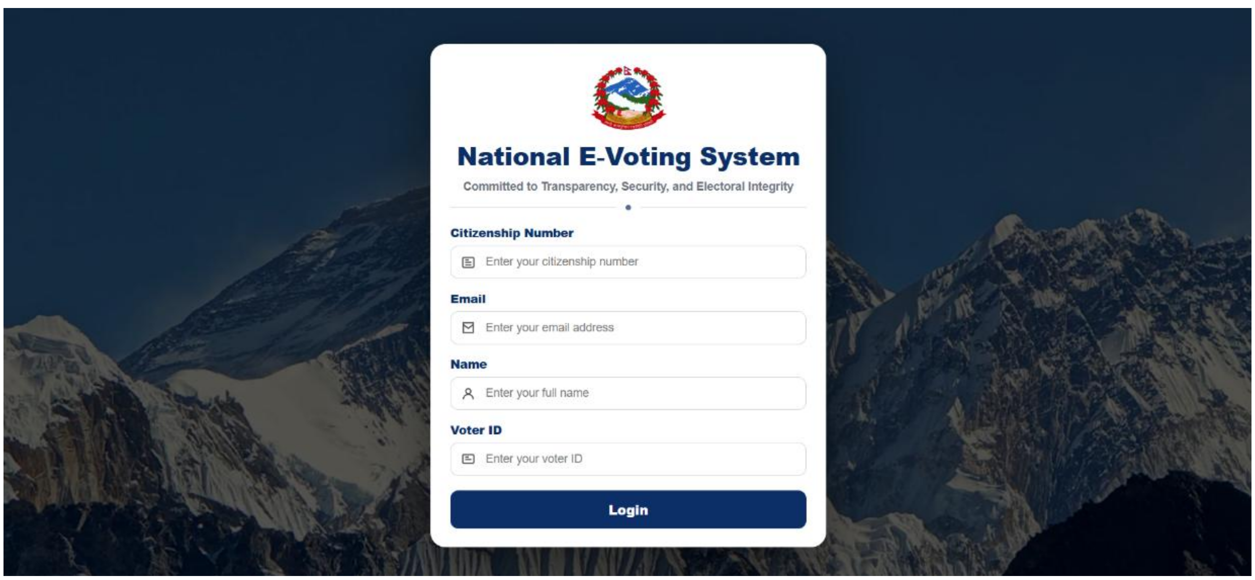


*Figure 9.Login Page*

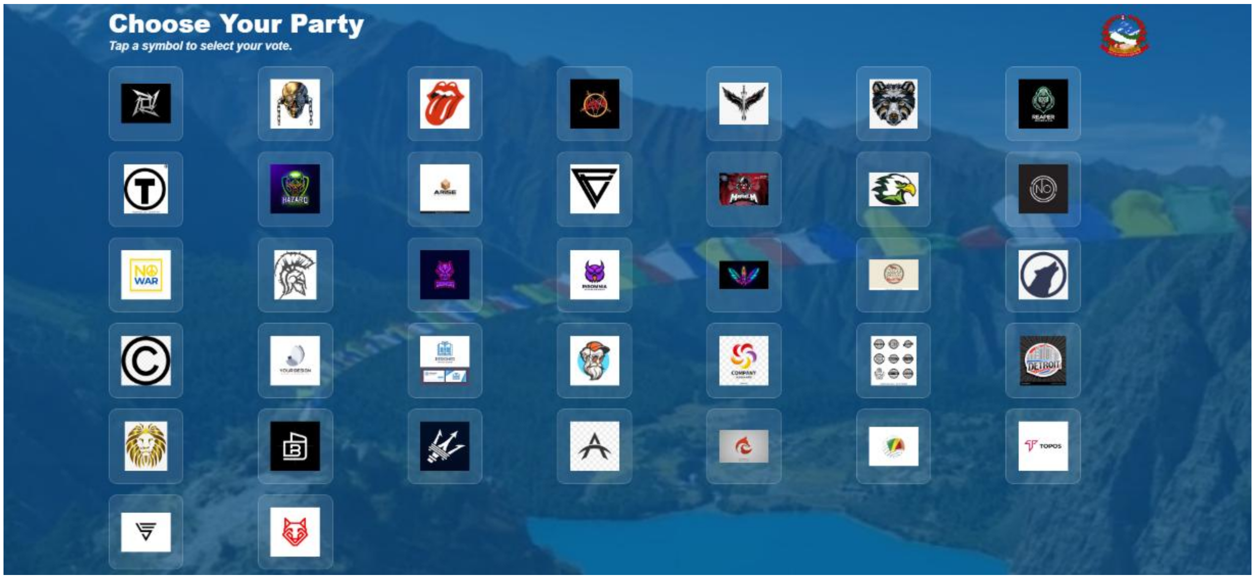


*Figure 10. Voting Page*


*Figure 11.Thankyou Page*

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